

1. Introduction:

Light significantly influences human emotion and behaviour. Circadian rhythms, sleep patterns, and mood can all be affected by different intensities and colours of light. This project explores the physiological effects of light on emotional states by analysing bioelectrical signals using non-invasive electrophysiology tools. The setup employs the **BioAmp EXG Pill** to acquire EEG/EMG/ECG data, an **ADS1115 analog-to-digital converter** for high-resolution signal sampling, and a **Raspberry Pi 3A+** for data processing and visualization. Our aim is to correlate variations in bio-signals with controlled light stimuli to interpret emotional responses.

We intend to implement our project in:

- To propose optimized lighting conditions for enhanced emotional well-being.
- Adaptive lighting that changes colour temperature and intensity based on user mood.
- Dynamic lighting in offices to minimize stress and increase efficiency.
- Light therapy for treating Seasonal Affective Disorder (SAD) and depression.
- Hospital lighting optimization to improve patient recovery rates and comfort.
- Mood-based lighting in stores to influence customer emotions and buying behaviour.
- Smart fitting room lighting that adapts to enhance clothing perception.

The novelty of your project lies in its integration of real-time emotional analysis with ambient light modulation using low-cost, compact hardware. Unlike conventional studies that rely on subjective self-reporting or expensive medical-grade systems, your setup utilizes non-invasive, accessible electrophysiological monitoring (BioAmp EXG Pill) and a Raspberry Pi-based processing unit to create a portable, scalable emotional-response detection system. Furthermore, your approach to correlating specific light stimuli (intensity and color) with EEG/EMG/ECG patterns provides a novel pathway for studying the neurophysiological basis of light-induced mood changes in real-world environments.

2. Literature Survey:

Serial No.	Paper Name	Major Details	Research Gap
1	Light Affects Mood and learning through Distinct Retina-Brain Pathways Author: Fernandez et	Proper light exposure is crucial for mental well being and cognitive health, with potential applications in light-based therapies for mood disorders like SAD and optimizing lighting environments.	Further research is needed to explore specific mechanisms and therapeutic applications of these pathways in diverse populations.
2	Light Affects Mood and Learning through Distinct Retina-Brain Pathways Author: Fernandez et al., 2018 (Cell Journal)	Proper light exposure is crucial for mental well-being and cognitive health, with potential applications in light-based therapies for mood disorders like SAD and optimizing lighting environments.	Further research is needed to explore specific mechanisms and therapeutic applications of these pathways in diverse populations.
3	Lighting Emotions: A Review of the Emotional Influence of Colour Perceived Lightness Author: AIC 2021 Conference (Specific authors not mentioned)	Lightness influences emotions, attention, memory, decision-making, and consumer behavior, with lighter colors being inviting and darker ones perceived as durable, highlighting the importance of optimized lighting design.	Future studies should eliminate methodological biases to refine the understanding of color-emotion relationships in various settings.
4	Lighting and Recognition of Emotion Conveyed by Facial Expressions Author: B. Yang and S. Fotios (School of Architecture, University of Sheffield, UK)	The study investigated how different lighting conditions affect the perception of emotions conveyed by facial expressions, finding that luminance follows a plateau–escarpment relationship, where performance improves with light levels up to a certain point but plateaus thereafter.	Further research is needed to explore the impact of factors like glare, luminance uniformity, and three-dimensional targets on facial expression recognition in real-world conditions.
5	Emotional Assessment and Detection Methods in Integrated Lighting Design for Enhanced Well-Being Author: Mayowa Abudu, Siegfried Yeboah, and Timothy Eccles (London South Bank University)	While subjective assessments offer insights, objective biometric methods improve precision, though challenges like environmental sensitivity and computational demands must be addressed for practical implementation.	Future research should advance sensor technology, AI models, and real-world applications to create fully functional emotion-responsive lighting systems.

6	The Impact of Daytime Light Exposures on Sleep and Mood in Office Workers Author: Mariana G. Figueiro, Bryan Steverson, Judith Heerwagen, Kevin Kampschroer, Claudia M. Hunter, Kassandra Gonzales, Barbara Plitnick, Mark S. Rea	The study measured personal circadian-effective light exposures in office workers across different seasons, demonstrating the positive impact of morning and daytime light on sleep quality and emotional well-being.	Further research is needed to optimize lighting conditions in office settings and explore long-term health benefits of increased circadian-effective light exposure.
7	The Potential Influence of LED Lighting on Mental Illness Author: Michael Bauer, Tasha Glenn, Scott Monteith, John F. Gottlieb, Philipp S. Ritter, John Geddes, and Peter C. Whybrow	LED lighting, especially with its dominant blue spectrum, affects non-image-forming (NIF) functions via melanopsin-sensitive retinal ganglion cells, influencing circadian rhythms, sleep regulation, and psychiatric conditions. The rapid global transition to LED lighting raises concerns about its unintended consequences on mental health.	More studies are needed to establish guidelines for safe LED lighting exposure, particularly for individuals with mental health conditions.
8	How Does Light Regulate Mood and Behavioural State? Author: Nina Milosavljevic	Blue light (~460 nm) is particularly effective in increasing alertness, reducing sleepiness, and influencing mood by directly stimulating ipRGCs.	The specific light properties (irradiance, colour, duration, and timing) that optimally regulate mood and behaviour are not yet fully understood.
9	Perceptual and Emotional Effects of Light and Colour in a Simulated Retail Space Author: Maria Lombana, Graciela Lucia Tonello	The study used a controlled experiment with different lighting directions (front, overhead, and wall washer) and four wall colour schemes (yellow-blue, magenta, grey, and green-magenta) to assess their effects on perception and emotions in retail spaces.	The study did not keep luminance constant across all colour conditions, making it difficult to generalize findings to other colours and lighting scenarios.
10	Light and Emotion: Achieving Emotions in Landscape Architecture by Using Light Author: Carmina Constanța Gheorghită, Mircea Grigorovschi, Dragoș-Emil Ciolacu-Miron	The study explores various lighting parameters—brightness, colour, direction, contrast, and time—and their effects on human emotions in outdoor spaces.	There is a lack of sufficient research on lighting in landscape architecture compared to traditional architectural lighting, necessitating deeper exploration of its emotional and perceptual impacts.
11	Dopamine and Light: Effects on Facial Emotion Recognition	The study tested women with subsyndromal seasonal affective disorder under	The study only included women, making it unclear whether the findings

	Author: Elizabeth Cawley, Maria Tippler, Nicholas J. Coupland, Chawki Benkelfat, Diane B. Boivin, Marije aan het Rot, Marco Leyton	bright (3000 lux) and dim (10 lux) light conditions after dopamine depletion, measuring their ability to recognize facial emotions.	generalize to men or individuals with more severe mood disorders.
12	Effect of Mood Lighting on Human Emotions and Behaviour Author: Dr. Rahul Suresh Honrao	The study used a mixed-methods approach, including surveys with 200 participants, to analyse emotional responses to different lighting conditions using statistical techniques like ANOVA and regression analysis.	The study did not explore long-term psychological effects of mood lighting, leaving room for further research on its sustained impact on well-being.
13	Developmental Changes in Emotion Recognition from Full-Light and Point-Light Displays of Body Movement Author: Patrick D. Ross, Louise Polson, Marie-Hélène Grosbras	The study used video and motion capture recordings of actors portraying emotions, testing recognition in 107 children (ages 4–17) and 14 adults under full-light and point-light conditions.	The study did not explore whether the observed developmental trajectory varies across different cultural or social contexts.
14	Effects of Light and Sound on the Prefrontal Cortex Activation and Emotional Function: A Functional Near-Infrared Spectroscopy Study Author: Shota Hori, Koichi Mori, Takehisa Mashimo, Akitoshi Seiyama	The study used functional near-infrared spectroscopy (fNIRS) on 757 subjects to assess how continuous mild stimuli affect brain activity and emotions, utilizing a real-time feedback system.	The study did not investigate how different cultural or environmental factors influence emotional responses to light and sound stimuli.
15	Effects of light on human circadian rhythms, sleep and mood Author: Christine Blume, Corrado Garbaza, Manuel Spitschan	Circadian rhythms, suprachiasmatic nuclei (SCN), artificial light, sleep-wake behavior, CRSWD, therapeutic light use.	Need for deeper understanding of the relationship between artificial light exposure and CRSWD development.
16	The Effects of Perceiving Colour in Living Environment on QEEG, Oxygen Saturation, Pulse Rate, and Emotion regulation in Humans	Light and colour effects, biological signals (SpO2, pulse rate, QEEG), emotional responses, BRUMS questionnaire, six colour environments.	Need for further exploration of physiological responses to colour in room design for both general and clinical settings.

	Author: Watchara Sroykham, Student member, IEEE J. Wongsathikun and Y. Wongsawat, Member, IEEE		
17	Recognizing emotions in faces: effects of acute tryptophan depletion and bright light Author: Marije aan het Rot, Nicholas Coupland, Diane B Boivin, Chawki Benkelfat1 and Simon N Young1	Acute tryptophan depletion (ATD), facial emotion perception, bright vs. dim light, crossover design, healthy women.	Influence of lighting conditions on ATD effects in emotion perception needs further exploration.
18	The Effect of Treatment room lighting colour on Time perception and Emotion Author: Seulki Han, Daehee Lee	Light colour, perceived treatment time, emotional reactions, ultrasound therapy, stopwatch method.	Need to mitigate negative mood effects of blue and red light in therapeutic settings.
19	The Effect of White Verses Coloured Light in Waiting Rooms on People's Emotion Author: Zhihui Zhang, Josep Maria Fort Mir, Lluís Gimenez Mateu	Architectural lighting, emotional impact, PAD model, white vs. coloured light, windowless waiting room.	Need for further investigation into effective lighting strategies for improving mood in non- clinical windowless spaces.
20	Effects of Illuminance and Correlated colour temperature on emotional responses and lighting adjustment behaviours Author: Armin Mostafavi, Tong Bill Xu, Sahel Kalantari	Lighting conditions, emotional response, adjustment behavior, gender effects, VR simulation	Need for personalized lighting design in indoor environments.

3. Working Principle:

a) Signal Acquisition:

The BioAmp EXG Pill captures electrophysiological signals (e.g., EEG for brain activity) from the subject under varying light conditions.

b) Signal Amplification and Conversion:

The raw analog signals are amplified by the BioAmp and digitized using the 16-bit ADS1115 ADC to ensure precision before sending to the Raspberry Pi.

c) Processing and Analysis:

The Raspberry Pi processes the digital signal to extract relevant features and correlates these with the type and intensity of light stimuli to infer emotional states.

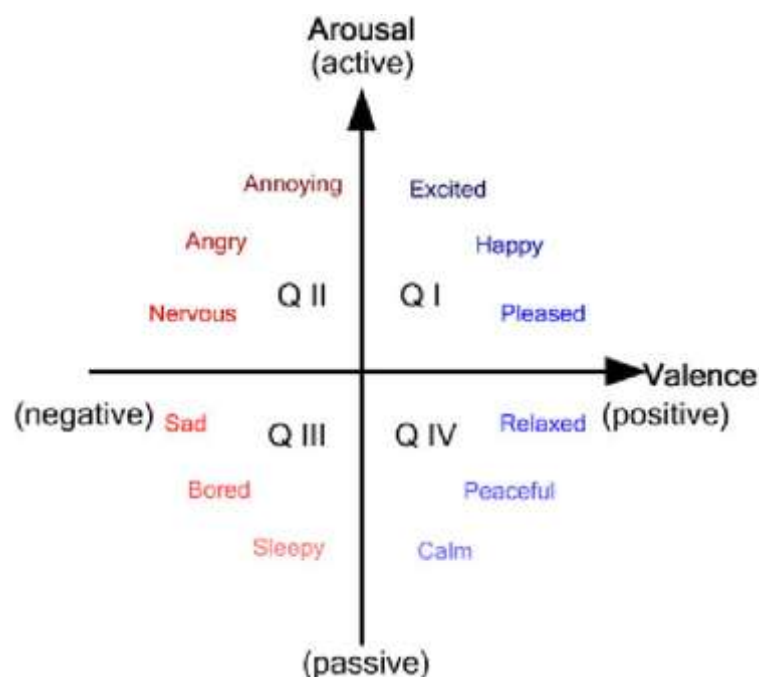
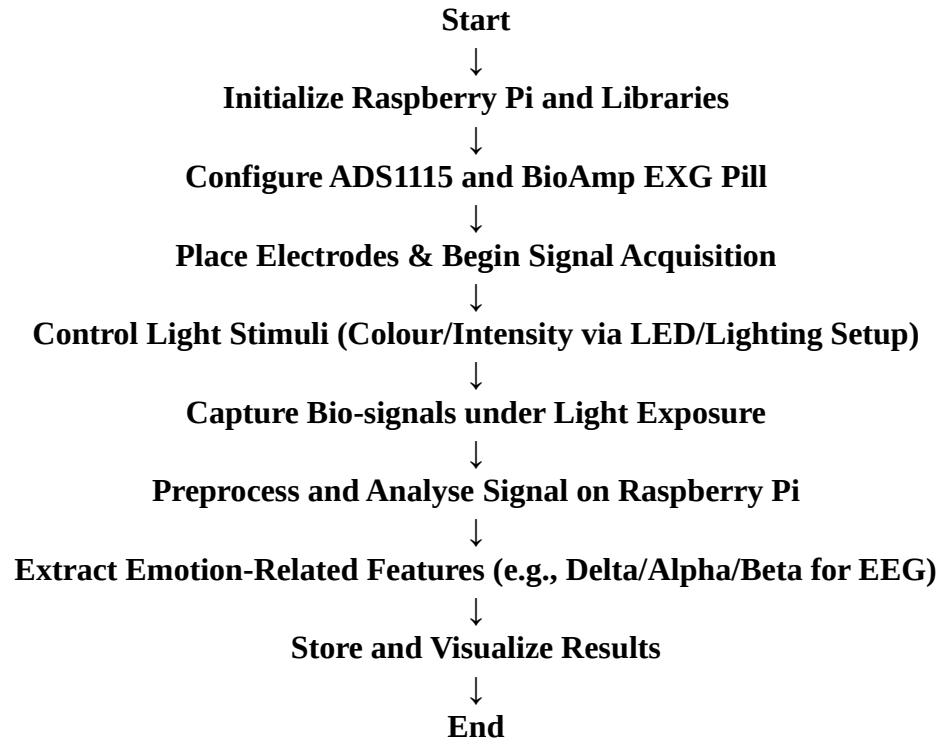


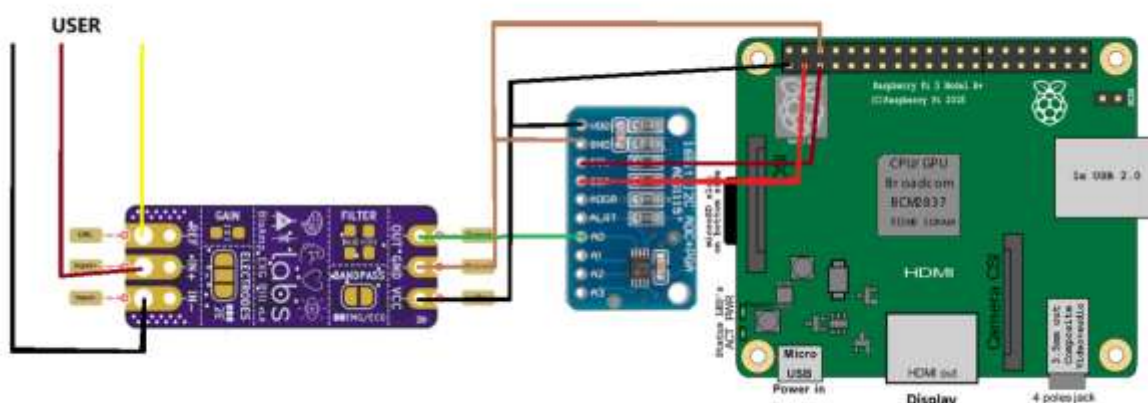
Fig 1: Arousal/Activation vs. Valence Emotional Spectrum

Our algorithm uses the digitized EEG signal to create an equivalent polar plot which is further divided into 12 equal sub-regions corresponding to each emotion. The algorithm calculates the percentage sample density on each section to formulate our results.

4. Flowchart:



5. Circuit Diagram:



BioAmp EXG Output	ADC A0
ADC SCL	Pin 5 - GPIO3 (SCL)
ADC SDA	Pin 3 - GPIO2 (SDA)

BioAmp EXG VCC	ADC VDD	Pin1/17 - +3.3V
BioAmp EXG GND	ADC GND	Pin 6 - GND

Fig 2: Circuit Diagram of our Prototype

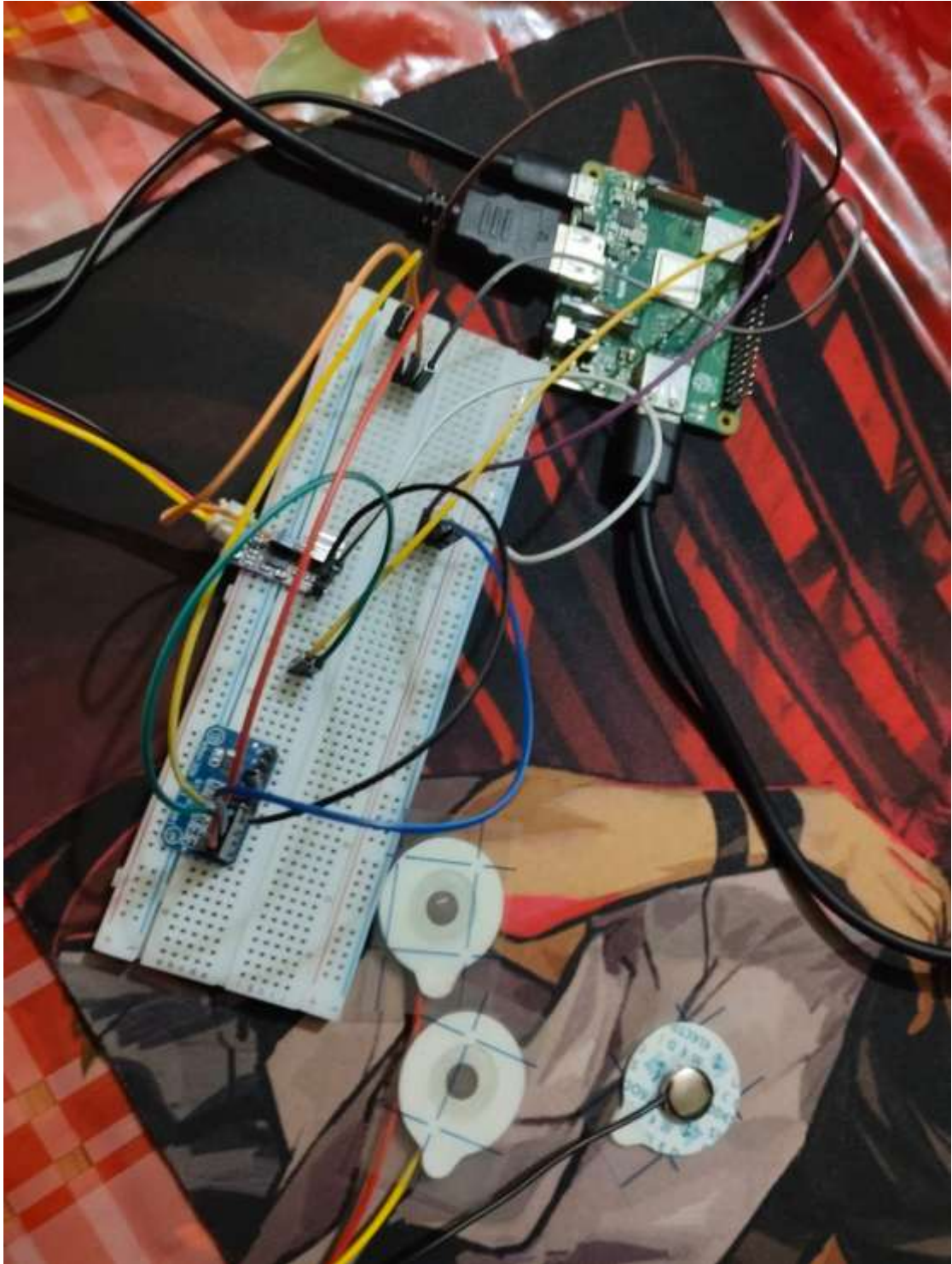


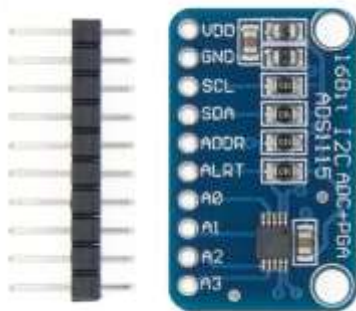
Fig 3: Prototype

6. Components:

COMPONENT	DESCRIPTION
BIOAMP EXG PILL	Analog front-end for bio-signal acquisition (EEG). Compact, open-source.
ADS1115	16-bit ADC with 4 channels and programmable gain; interfaces via I2C.
RASPBERRY PI 3A+	Small-form-factor Linux computer; handles signal processing, I2C control, and UI.
ELECTRODES	Non-invasive gel electrodes for signal acquisition from the skin surface.
LIGHT SOURCE	White/RGB LED or bulb to simulate different light environments.
WIRES & BREADBOARD	Standard prototyping hardware for connections.

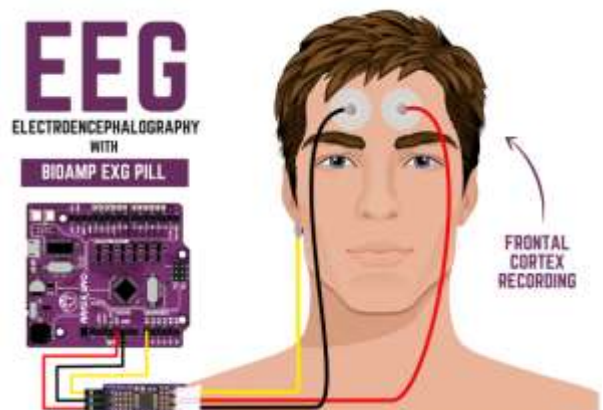


Raspberry Pi 3A+ is used to process the signal via Python and give us the output.



ADS1115 Analog to Digital Converter is used to convert the analog output of the EEG signal into a digital bitstream to be read by the Raspberry Pi.

EXG Pill reads the EEG of the user as an analog signal.



7. Results and Inference:

- Our prototype was run with two survey participants.
- Each participant was subjected to three lighting conditions.
- The following results were obtained:

1) Participant 1:

<i>Colour of Light/ Mood</i>	White (Control)	Red	Green
Pleased	7.4%	8%	7%
Happy	7%	7.6%	10%
Excited	8.2%	8.8%	6.8%
Annoyed	6.2%	7.4%	8%
Angry	9%	9.2%	7.4%
Nervous	6.6%	6.6%	10.2%
Sad	9.6%	9.2%	7.2%
Bored	8.6%	8.6%	9%
Sleepy (Tired)	10.4%	9.2%	10.2%
Calm	10.4%	8%	7.8%
Peaceful	9.6%	9.8%	8.6%
Relaxed	7%	7.6%	7.8%

2) Participant 2:

<i>Colour of Light/ Mood</i>	White (Control)	Red	Green
Pleased	8.5%	3.5%	4.75%
Happy	7.25%	7%	5.25%
Excited	6%	8.25%	9.5%
Annoyed	5.5%	6.75%	8%
Angry	9.75%	9%	7.25%
Nervous	10%	10.5%	6%
Sad	4%	5.5%	10%
Bored	11%	11.25%	11.5%
Sleepy (Tired)	8%	8%	8.25%
Calm	6.5%	6%	10.25%
Peaceful	12%	12%	9%
Relaxed	11.5%	12.25%	10.25%

```

srijon@raspberrypi:~$ python3 emotionpercentg.py
collecting 500 samples.....
EEG signal Emotion Mapping
pleased :37 samples(7.400000%)
happy :35 samples(7.000000%)
excited :41 samples(8.200000%)
annoyed :31 samples(6.200000%)
angry :45 samples(9.000000%)
nervous :33 samples(6.600000%)
sad :48 samples(9.600000%)
bored :43 samples(8.600000%)
sleepy :52 samples(10.400000%)
calm :52 samples(10.400000%)
peaceful :48 samples(9.600000%)
relaxed :35 samples(7.000000%)
srijon@raspberrypi:~$

```

Person 1: Test case white

```

srijon@raspberrypi:~$ python3 emotionpercentg.py
collecting 500 samples.....
EEG signal Emotion Mapping
pleased :40 samples(8.000000%)
happy :38 samples(7.600000%)
excited :44 samples(8.800000%)
annoyed :37 samples(7.400000%)
angry :46 samples(9.200000%)
nervous :33 samples(6.600000%)
sad :46 samples(9.200000%)
bored :43 samples(8.600000%)
sleepy :46 samples(9.200000%)
calm :46 samples(9.200000%)
peaceful :49 samples(9.800000%)
relaxed :38 samples(7.600000%)
srijon@raspberrypi:~$

```

Person 1: Test case red

```

srijon@raspberrypi:~$ python3 emotionpercentg.py
collecting 500 samples.....
EEG signal Emotion Mapping
pleased :35 samples(7.000000%)
happy :50 samples(10.000000%)
excited :34 samples(6.800000%)
annoyed :40 samples(8.000000%)
angry :37 samples(7.400000%)
nervous :51 samples(10.200000%)
sad :36 samples(7.200000%)
bored :45 samples(9.000000%)
sleepy :51 samples(10.200000%)
calm :39 samples(7.800000%)
peaceful :43 samples(8.600000%)
relaxed :39 samples(7.800000%)
srijon@raspberrypi:~$

```

Person 1: Test case green

Thus, we can infer that white light leads to more mellowed emotional balance, whereas red and green are leading to more intense emotional states.

8. Conclusion:

This project demonstrates a novel, low-cost approach to studying the impact of light on human emotion through real-time analysis of bioelectrical signals. By integrating the BioAmp EXG Pill, ADS1115 ADC, and Raspberry Pi 3A+, we successfully correlated light stimuli with physiological responses. The system offers a portable, non-invasive solution for monitoring emotional states, paving the way for applications in mental health, adaptive lighting, and human-centric design. Our findings highlight the potential of accessible embedded systems in bridging neuroscience and environmental interaction, encouraging further exploration into emotion-aware technologies.

9. Future Scope of Application:

- **Smart Lighting Systems:** Integration with home automation to adjust lighting based on real-time mood detection.
- **Mental Health Monitoring:** Continuous passive tracking of emotional wellbeing using light-stimulus responses.
- **Workplace Optimization:** Dynamic lighting adjustments to maximize productivity and reduce stress.
- **VR/AR Integration:** Enhance immersive experiences with mood-based environmental lighting.
- **Clinical Studies:** Further research on seasonal affective disorder (SAD) and therapeutic light exposure.

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